Stats Modelling II Project - EDA + Initial Modelling

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```
rm(list = ls())

library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.2
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
## Warning: package 'stringr' was built under R version 4.5.2
```

```
## Warning: package 'forcats' was built under R version 4.5.2
```

```
## Warning: package 'lubridate' was built under R version 4.5.2
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.2.0     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.0     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```r
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.2
```

```r
library(survey)
```

```
## Warning: package 'survey' was built under R version 4.5.3
```

```
## Loading required package: grid
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loading required package: survival
##
## Attaching package: 'survey'
##
## The following object is masked from 'package:graphics':
##
##     dotchart
```

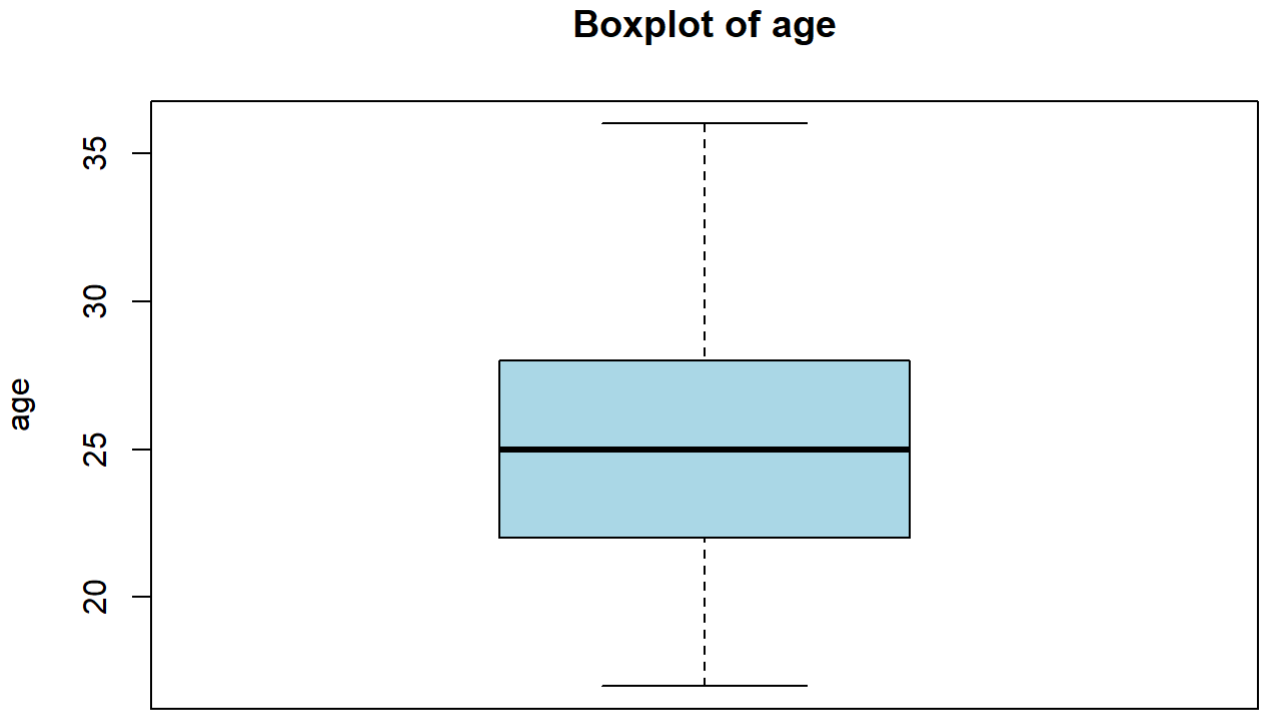
```r
library(ggplot2)
library(dplyr)
library(labelled)
```

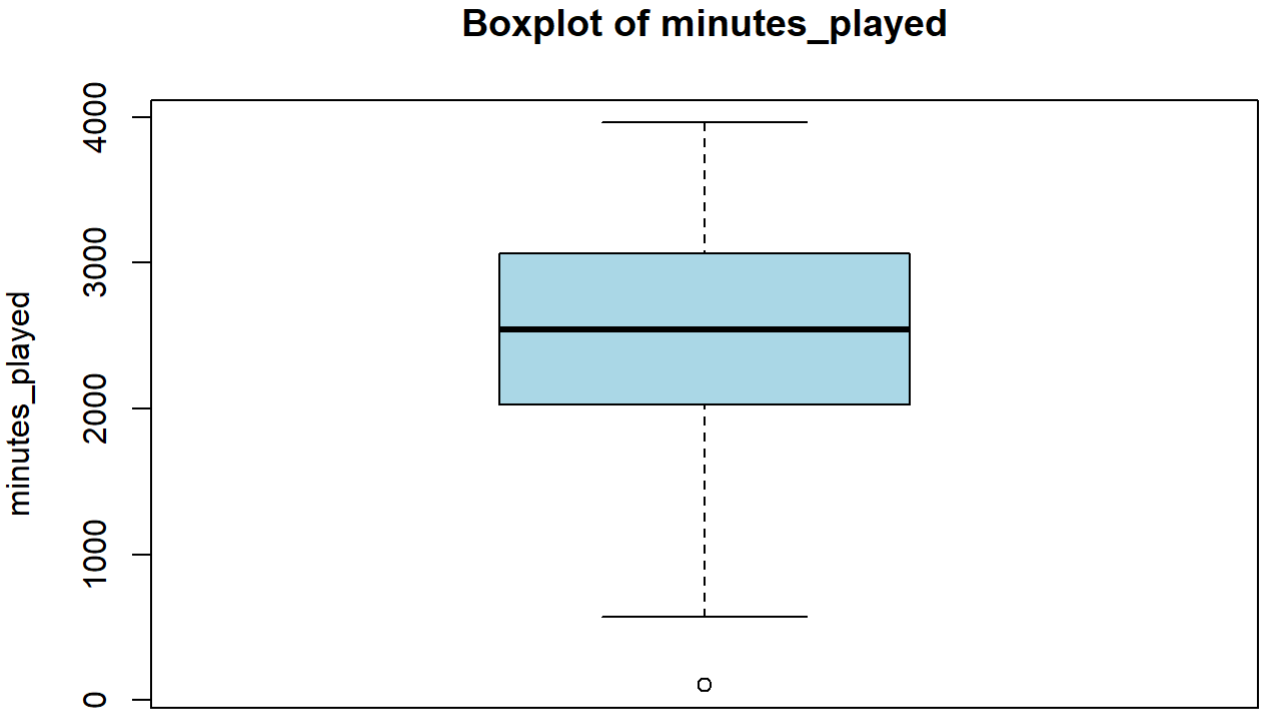
```
## Warning: package 'labelled' was built under R version 4.5.3
```

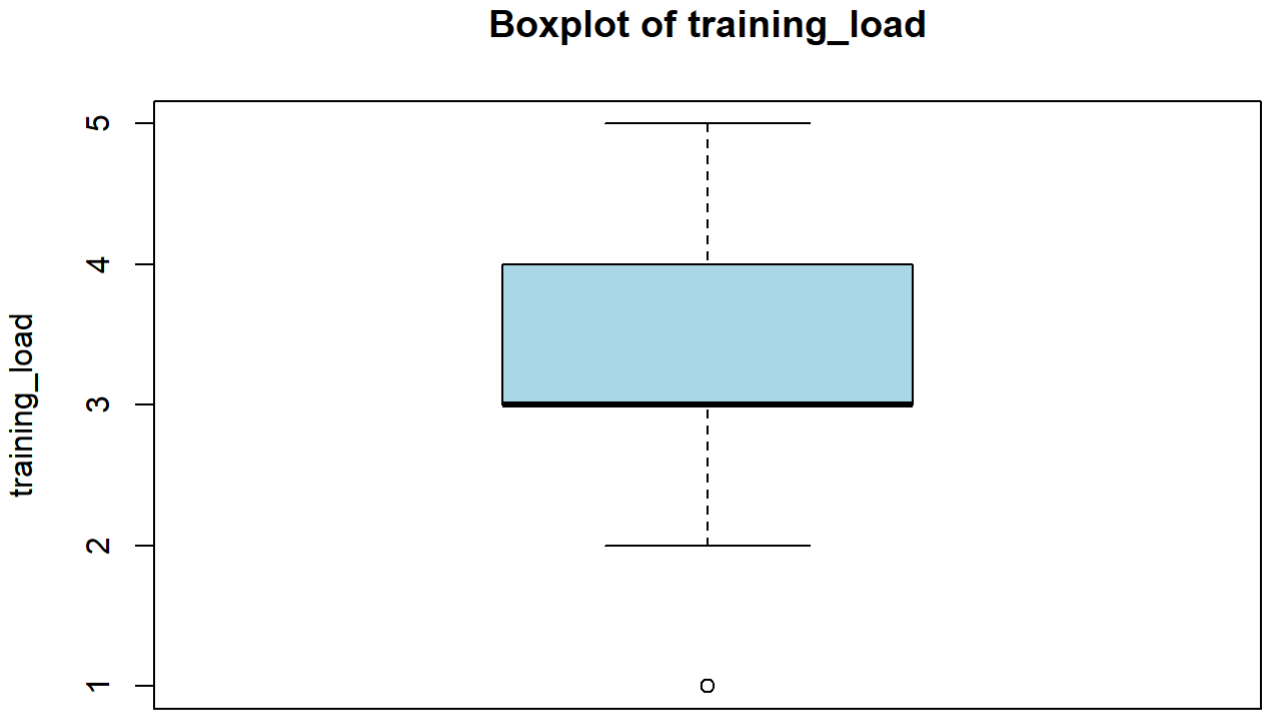
```r
matches = read.csv('ashford_matches.csv')
players = read.csv('league_players.csv')
```

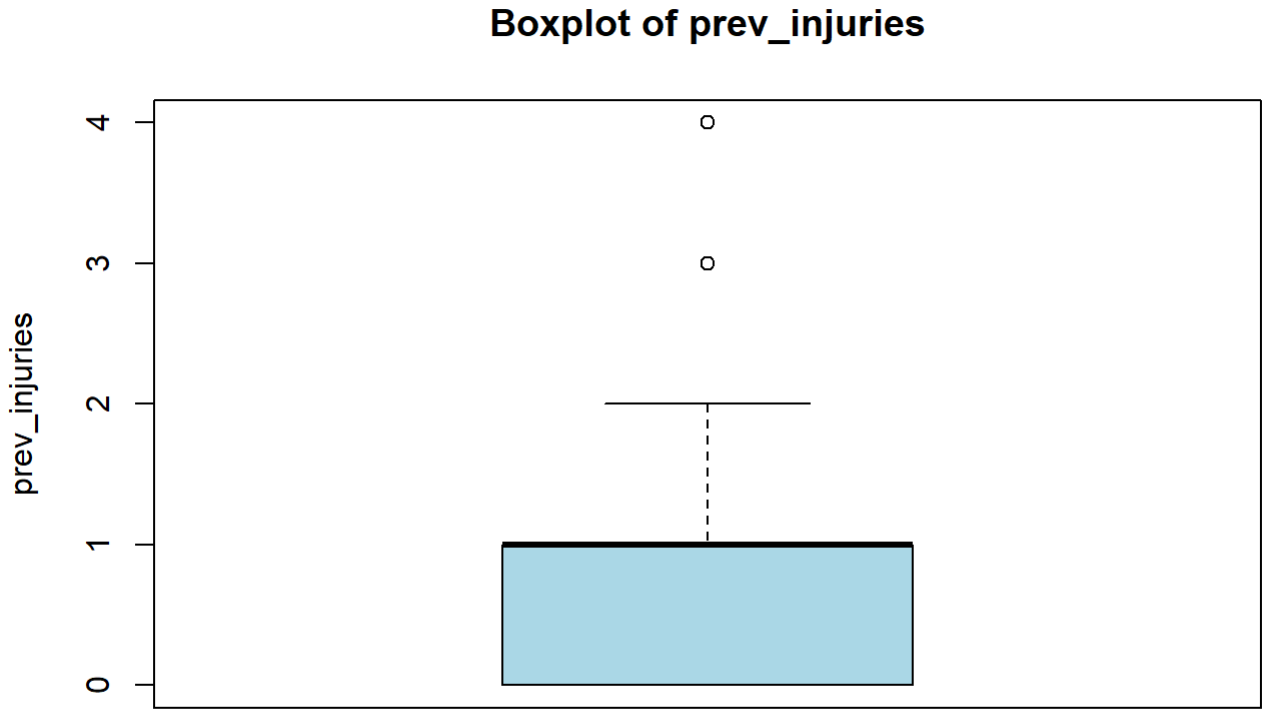
```r
# Select numeric variables
numeric_data <- players[sapply(players, is.numeric)]

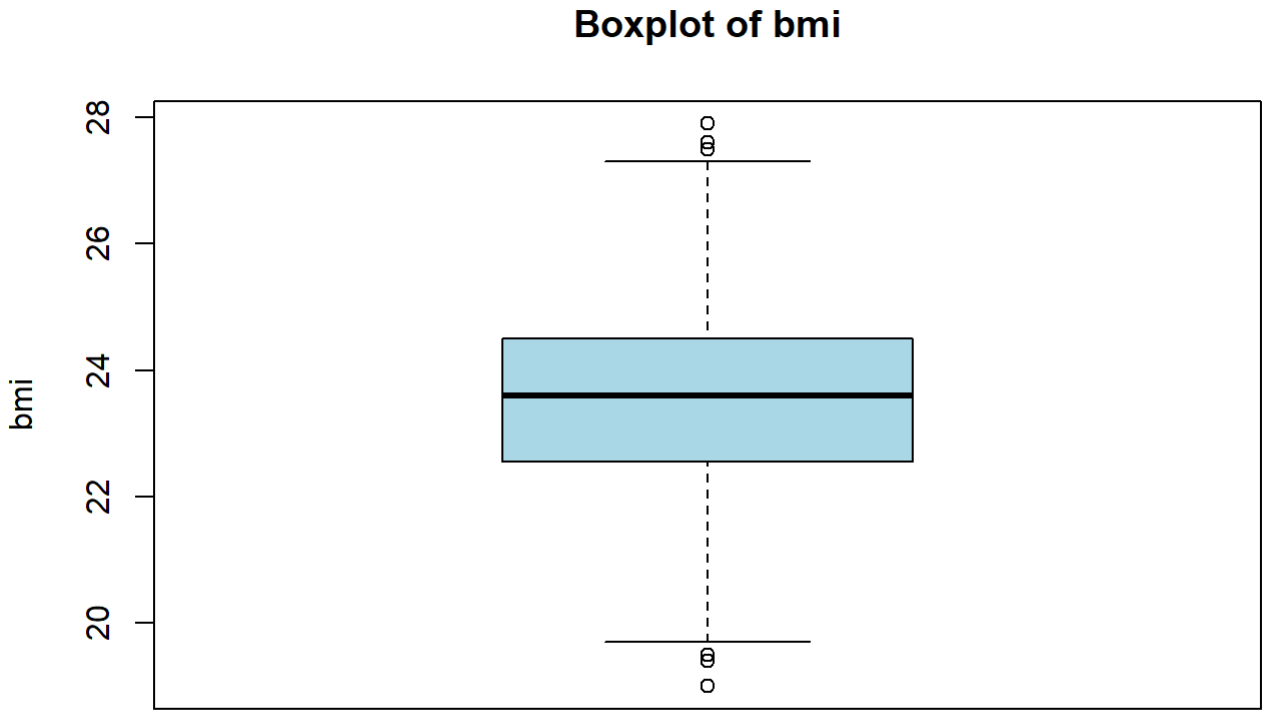
# Create a boxplot for each numeric variable
for (var in names(numeric_data)) {
  boxplot(numeric_data[[var]],
          main = paste("Boxplot of", var),
          ylab = var,
          col = "lightblue")
}
```

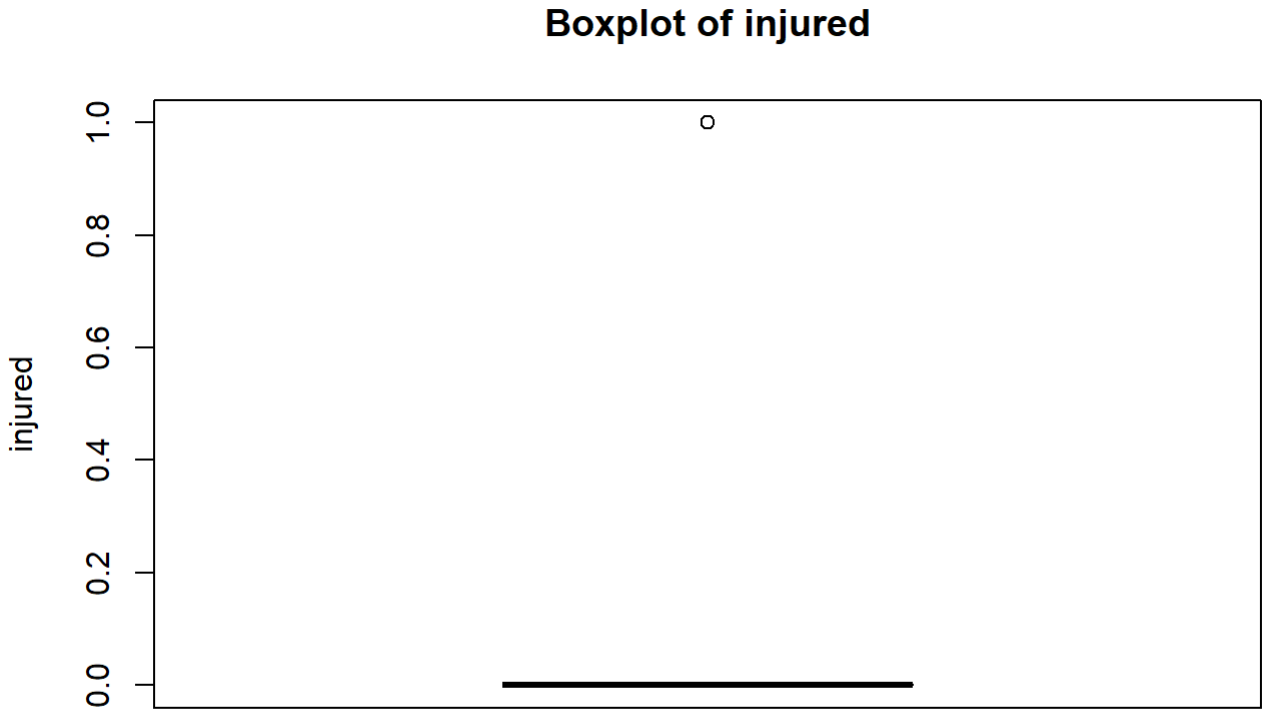

Boxplot of age
age
20
25
30
35


Boxplot of minutes_played
minutes_played
0
1000
2000
3000
4000


Boxplot of training_load
training_load
1
2
3
4
5


Boxplot of prev_injuries
prev_injuries
0
1
2
3
4

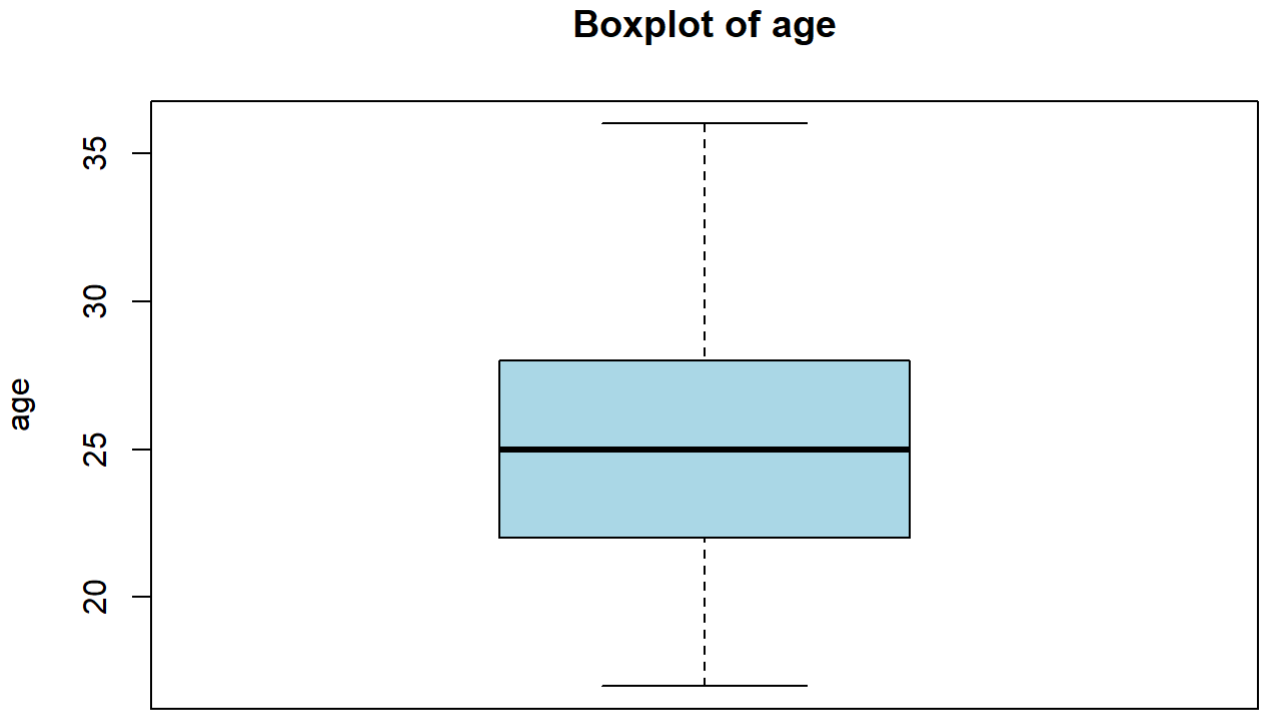

Boxplot of bmi
bmi
20
22
24
26
28

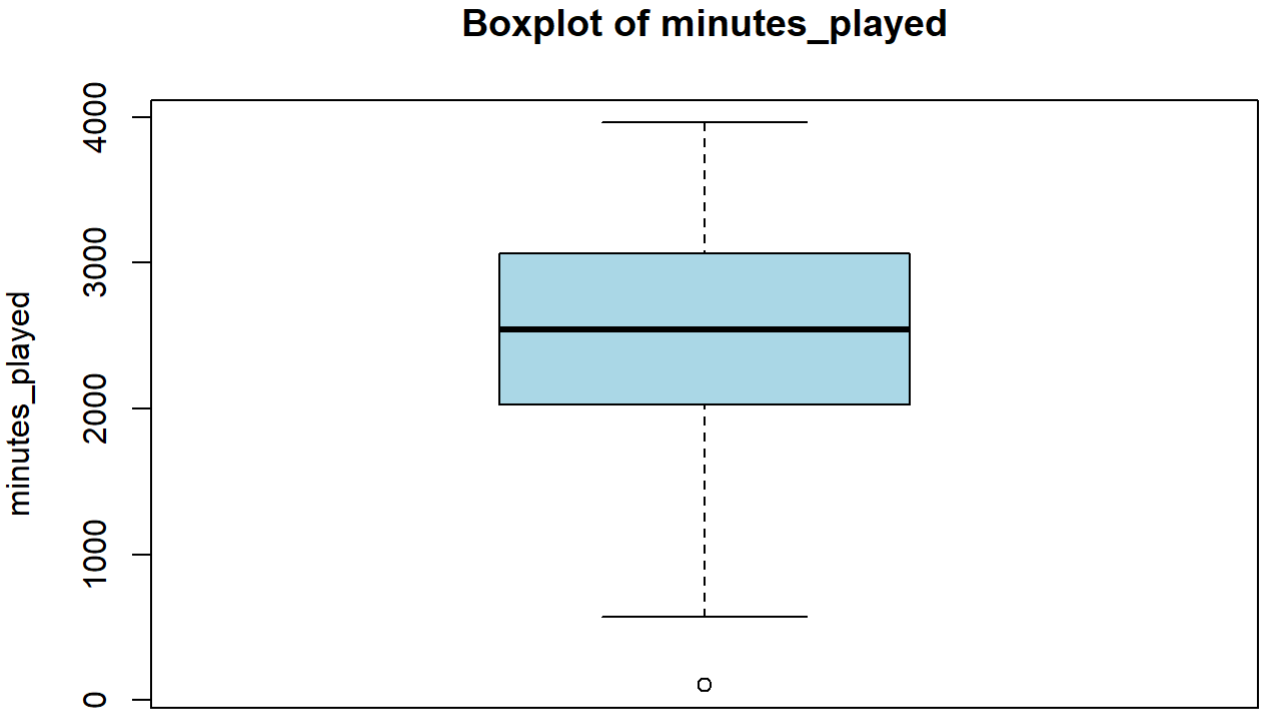

Boxplot of injured
injured
0.0
0.2
0.4
0.6
0.8
1.0

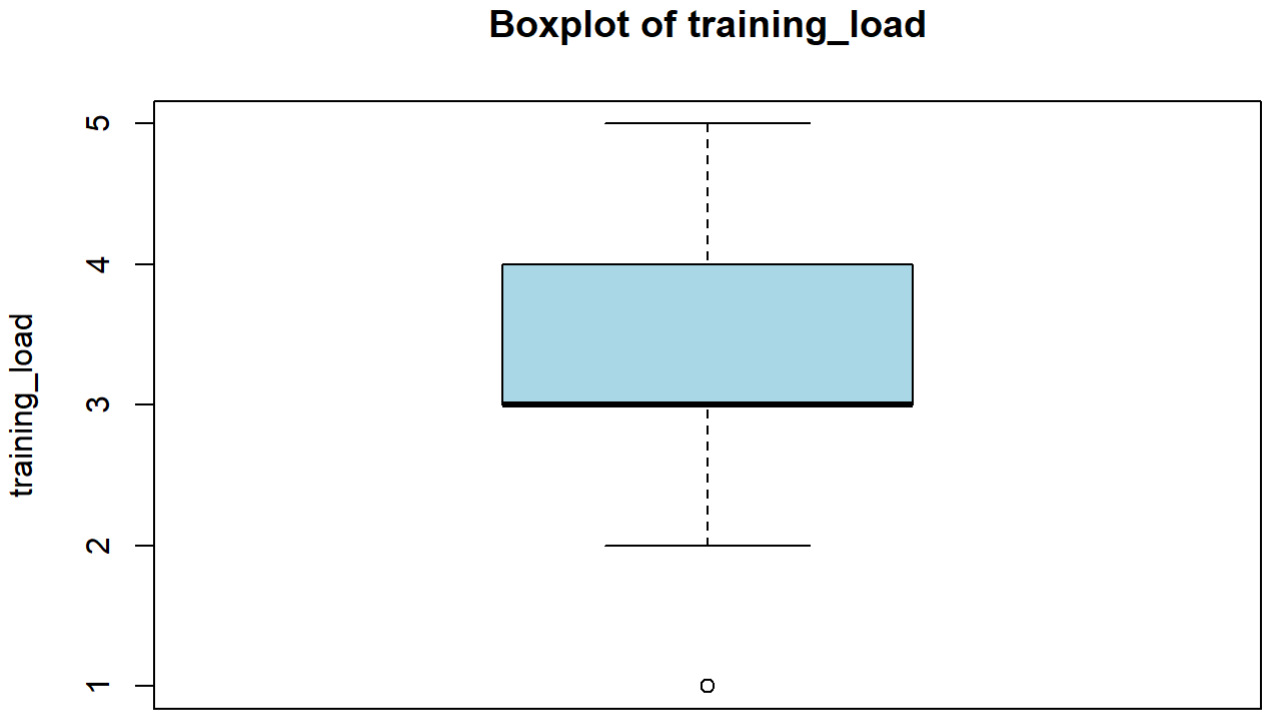
```r
# Create a dummy variable to represent home_away as a binary numeric variable

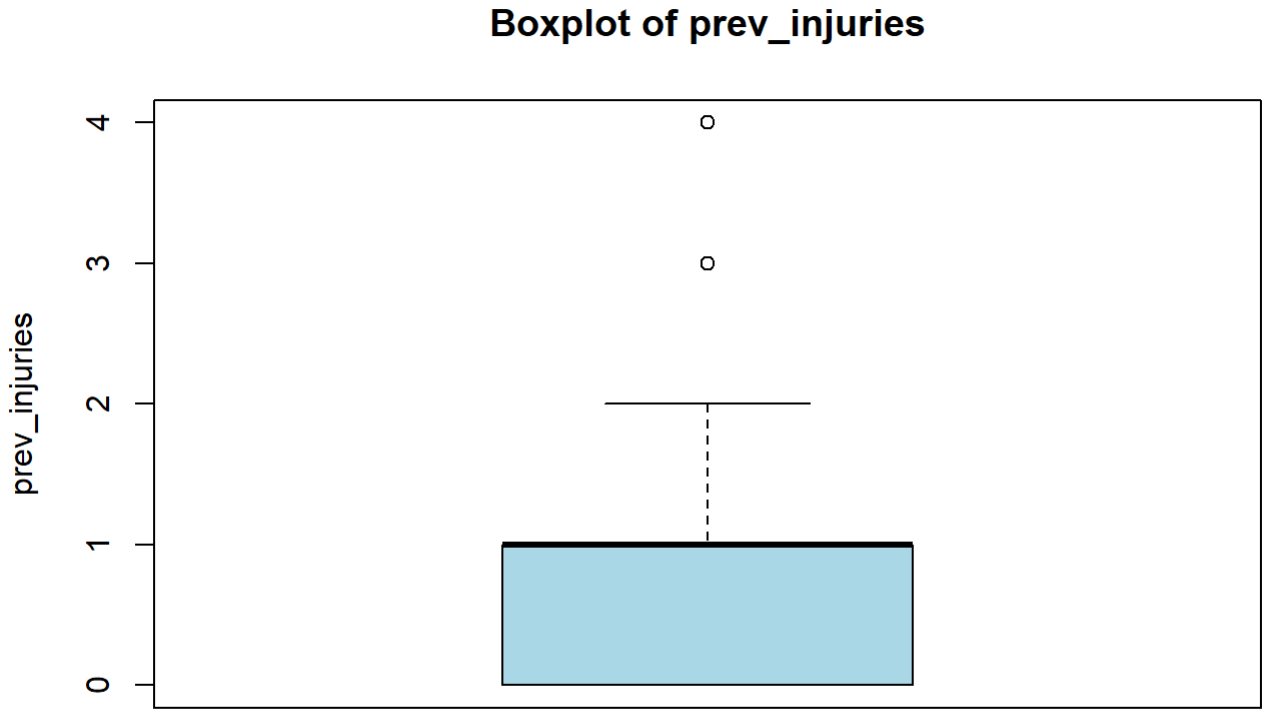
# Select numeric variables
numeric_data <- players[sapply(players, is.numeric)]

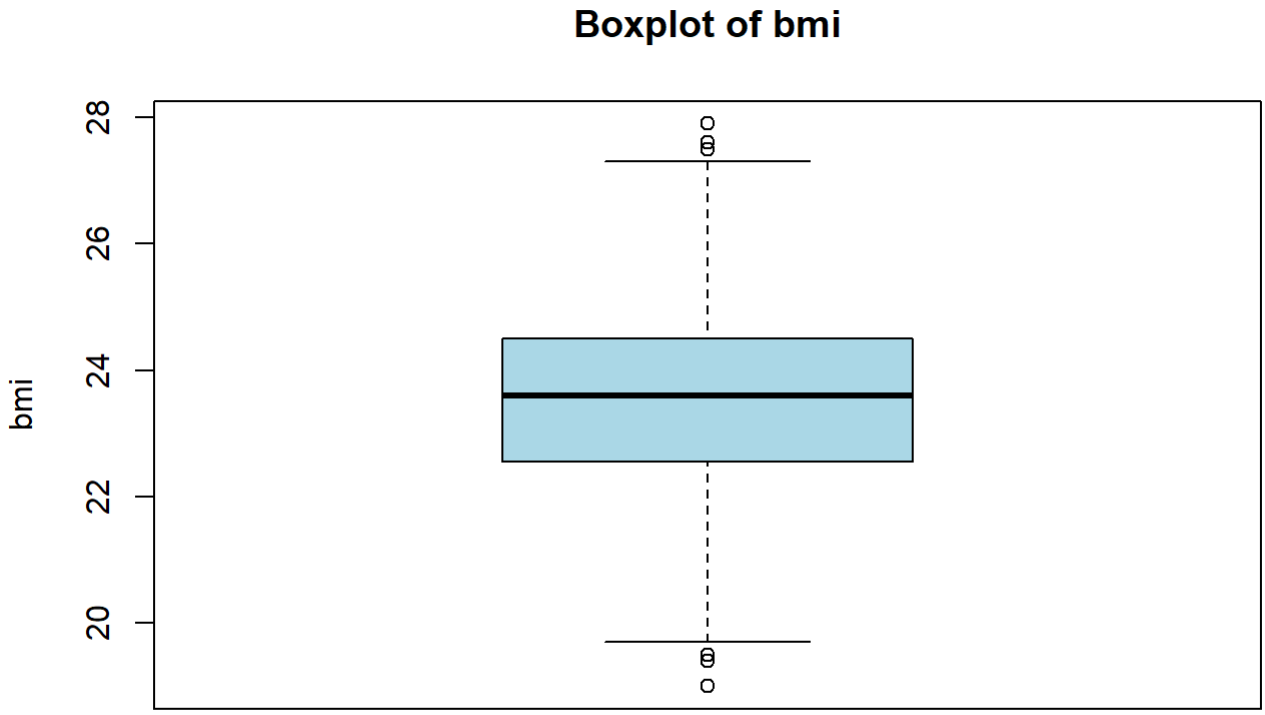
# Create a boxplot for each numeric variable
for (var in names(numeric_data)) {
  boxplot(numeric_data[[var]],
          main = paste("Boxplot of", var),
          ylab = var,
          col = "lightblue")
}
```

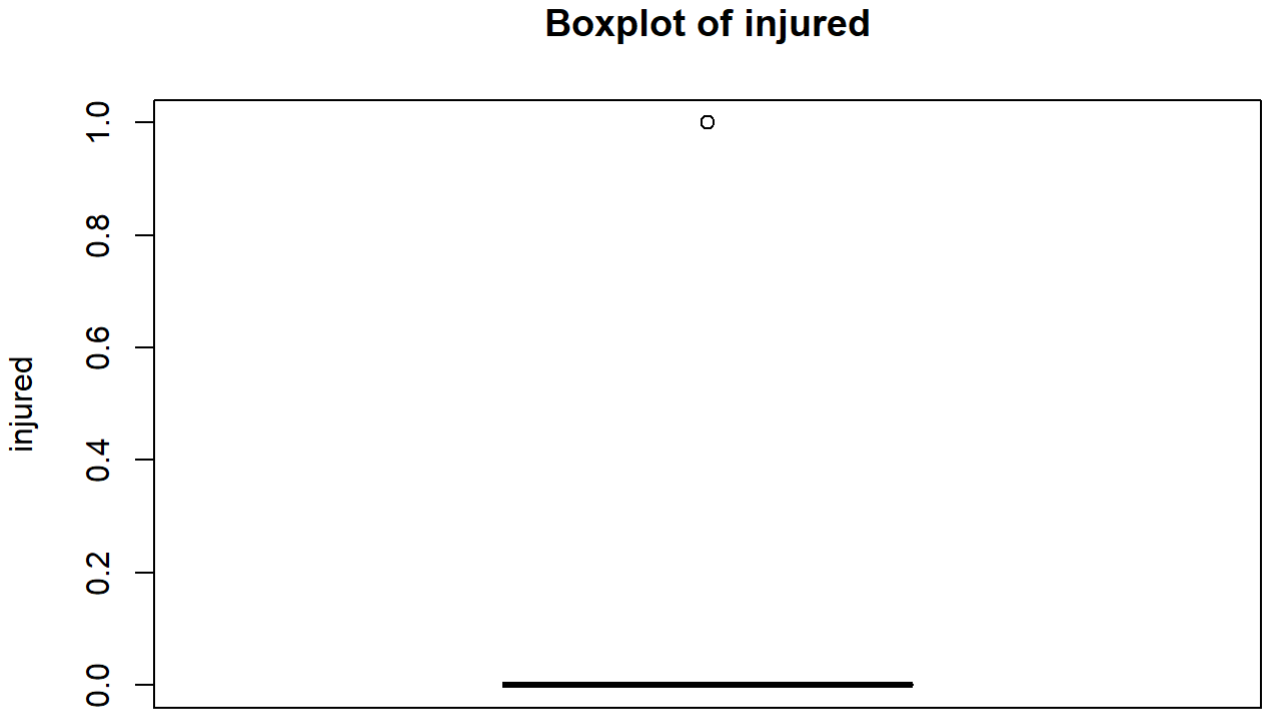

Boxplot of age
age
20
25
30
35


Boxplot of minutes_played
minutes_played
0
1000
2000
3000
4000

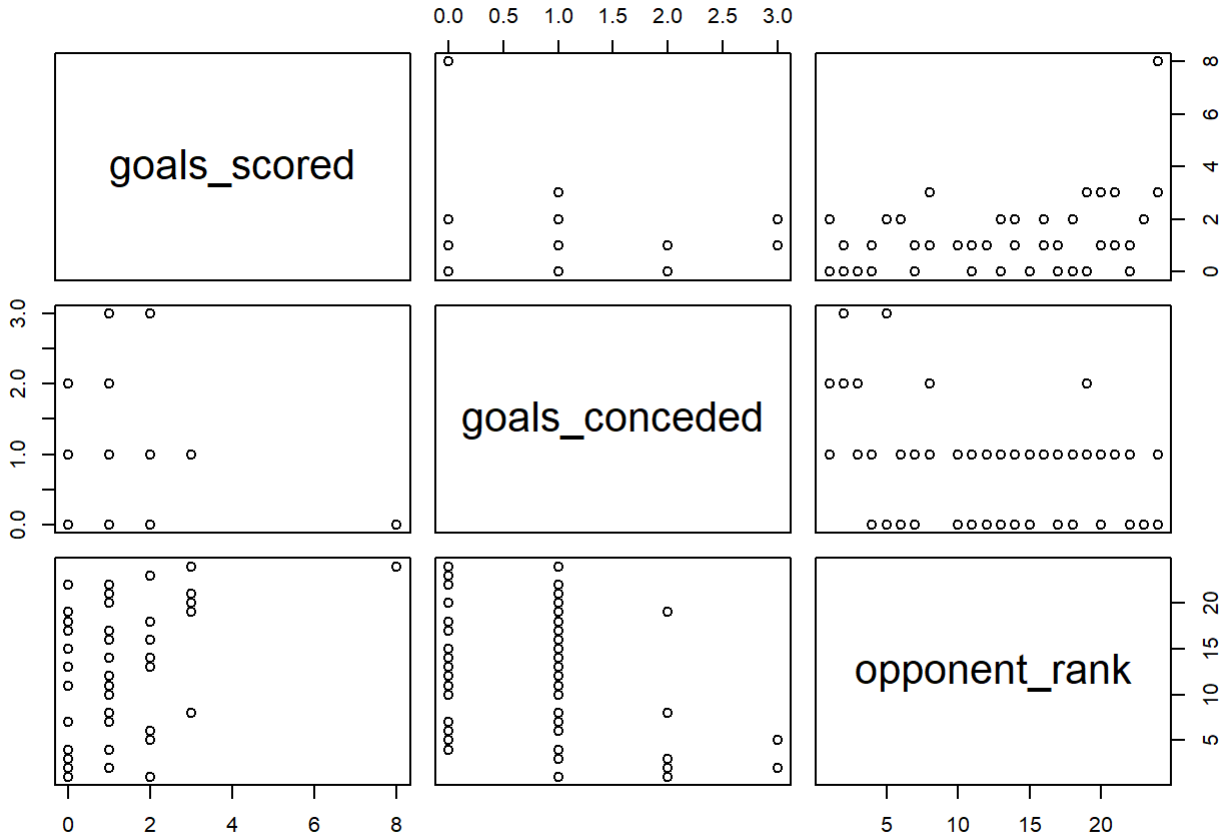

Boxplot of training_load
training_load
1
2
3
4
5


Boxplot of prev_injuries
prev_injuries
0
1
2
3
4

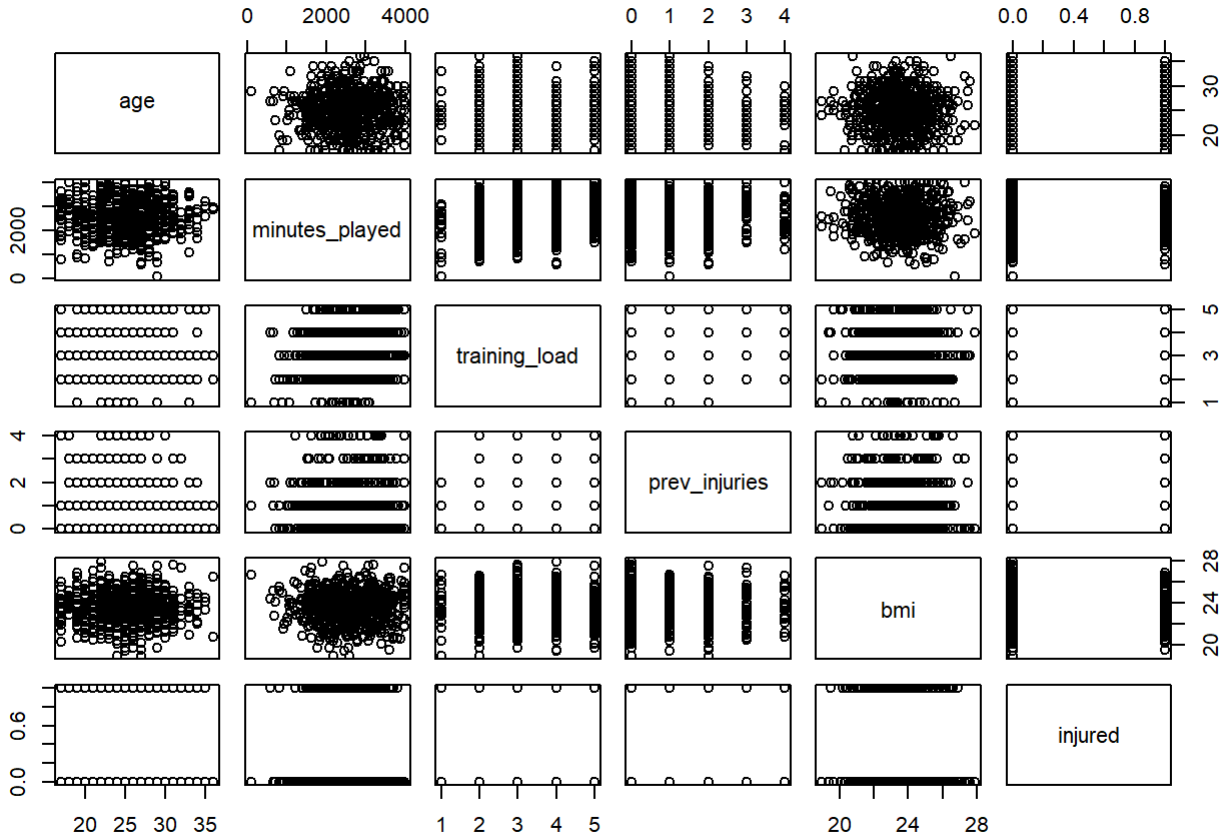





```
pairs(matches[, 2:4])
```




```
pairs(players[, 3:8])
```



In the ‘Matches’ dataset, there seems to be some relation between goals_scored/opponent_rank and goals_conceded/opponent_rank. We investigate this further below by fitting a Normal Linear Model to the data with opponent_rank as the response:

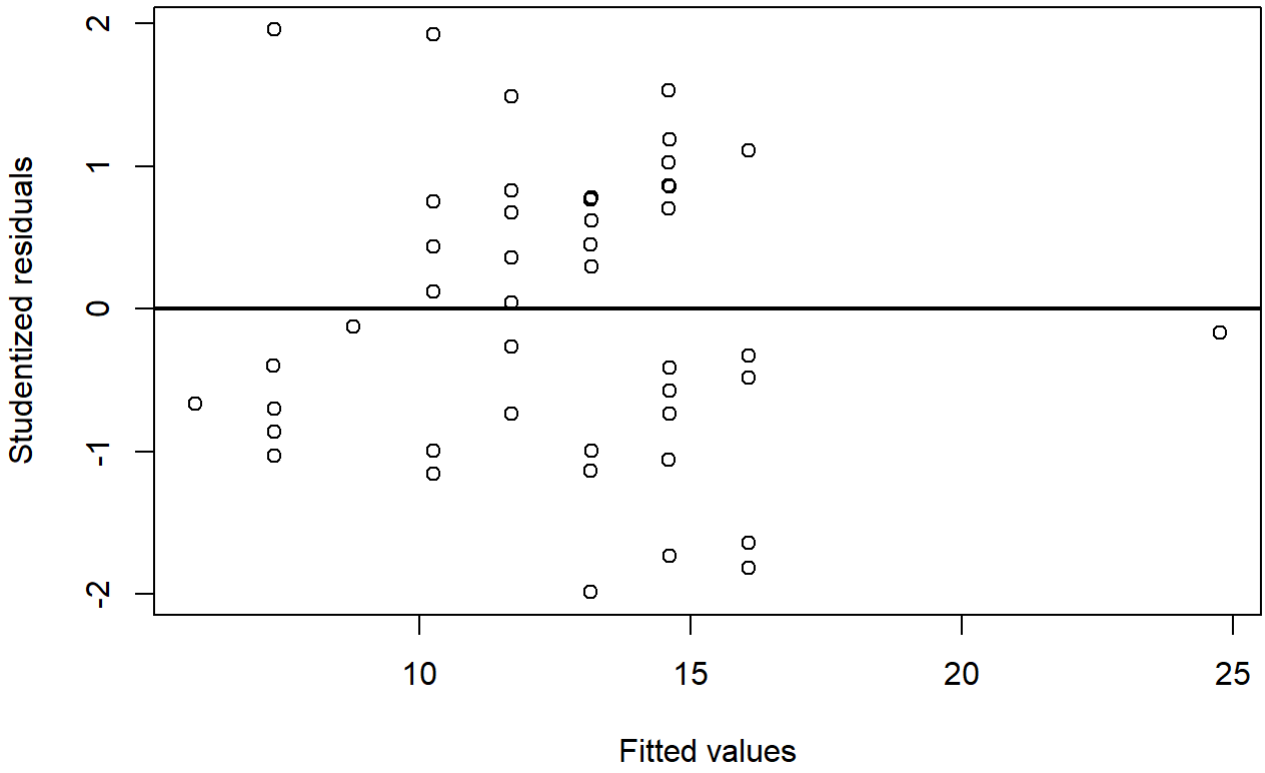
```
model1 = lm(opponent_rank~goals_scored+goals_conceded, data=matches)

summary(model1)
```

```
##
## Call:
## lm(formula = opponent_rank ~ goals_scored + goals_conceded, data = matches)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.1523  -4.6834  -0.2286   4.8428  11.7429
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     13.1718     1.7365   7.585 1.85e-09 ***
## goals_scored     1.4476     0.6874   2.106   0.0411 *
## goals_conceded  -2.9146     1.2174  -2.394   0.0211 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.418 on 43 degrees of freedom
## Multiple R-squared:  0.2213, Adjusted R-squared:  0.1851
## F-statistic: 6.111 on 2 and 43 DF,  p-value: 0.004617
```

We observe that both goals_scored and goals_conceded are significant to the 5% level. Now we plot residuals and check the Q-Q plot:

```
plot(model1$fitted, rstudent(model1), xlab='Fitted values', ylab='Studentized residuals')
abline(h=0, lwd=2)
```




```
qqnorm(rstudent(model1), main=NULL)
qqline(rstudent(model1), lwd=2)
```

